

Homework Assignment 2A

PWM Feedback Loops

ET4382 - Digital IC Design

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Part 1

1st Order Feedback Loop

Part 1: Circuit Description

1st order self-oscillating PWM feedback loop at ~1MHz

Signal flow:

Input current I_{in} injected at virtual ground of RC-integrator

Integrator (limiting_diffamp, gain=1e6, output limited to +/-2V)

Subtractor computes: integrator output - reference (gnd)

Hysteresis block creates PWM switching at +/-10V

VPWM feeds back through RFB to virtual ground (closing the loop)

VPWM drives LC filter + 4 Ohm load

Key components:

RFB = 100 kOhm, CINT = 250 pF (IC = 1V)

Hysteresis: sigout_high=10, sigout_low=-10, tdel=1ns

LC filter: L = 18 uH, C = 820 nF, R = 4 Ohm

Part 1: CINT Calculation & Transimpedance

Large-signal stability criterion:

$$CINT \geq VPWM * T / (2 * RFB * Vclip)$$

$$CINT \geq 10 * 1\mu s / (2 * 100k * 2V) = 250 \text{ pF}$$

Integrator unity-gain bandwidth:

$$fUGB = 1 / (2\pi * RFB * CINT) = 6.37 \text{ kHz}$$

Step 2: 1kHz current input (10 uA):

$$Vout = Iin * RFB = 10 \text{ uA} * 100 \text{ kOhm} = 1V \text{ (confirmed by simulation)}$$

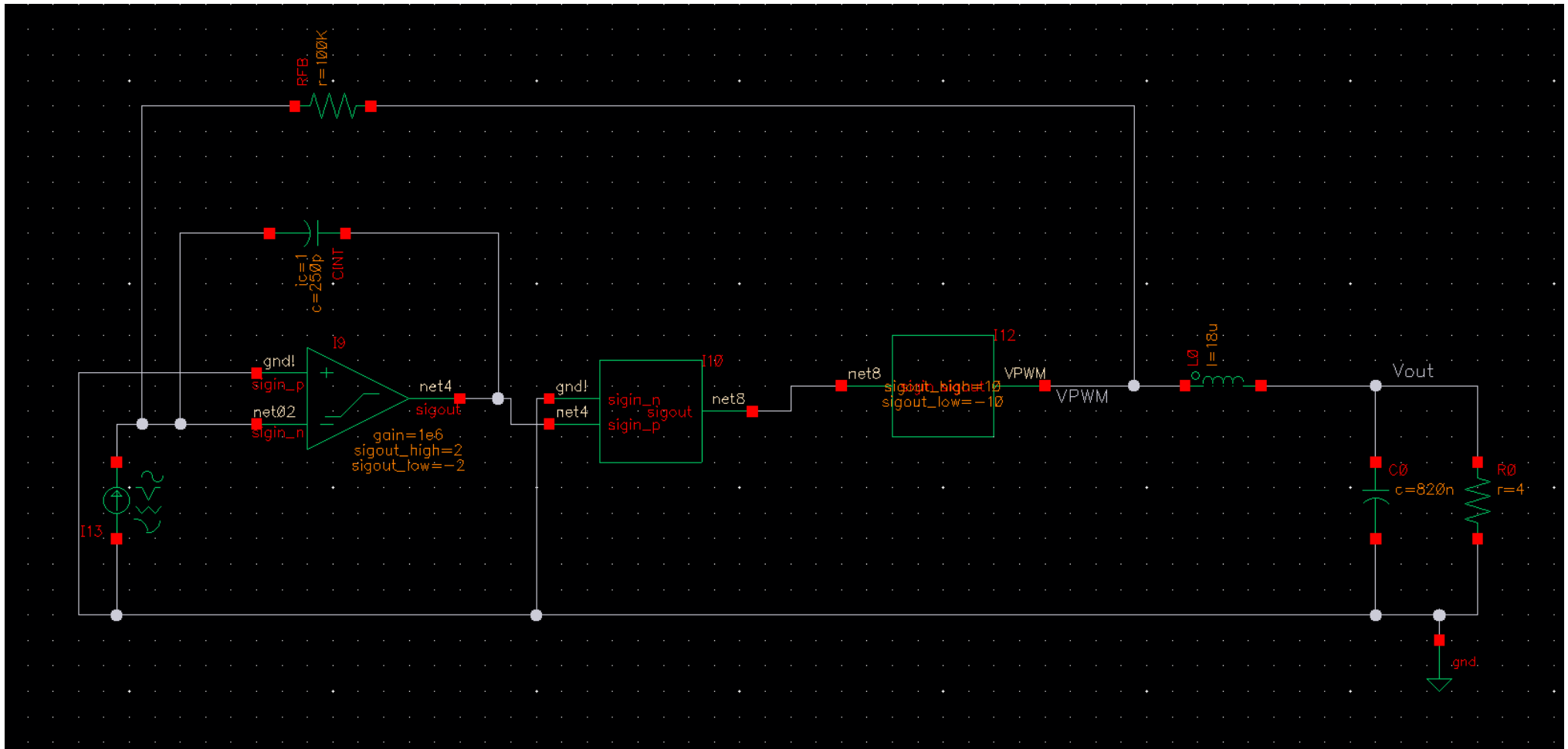
Clean sinusoidal output, 1kHz peak at ~0 dB in spectrum

IC = 1V on CINT is critical:

Hysteresis is edge-triggered, needs initial condition to start oscillation

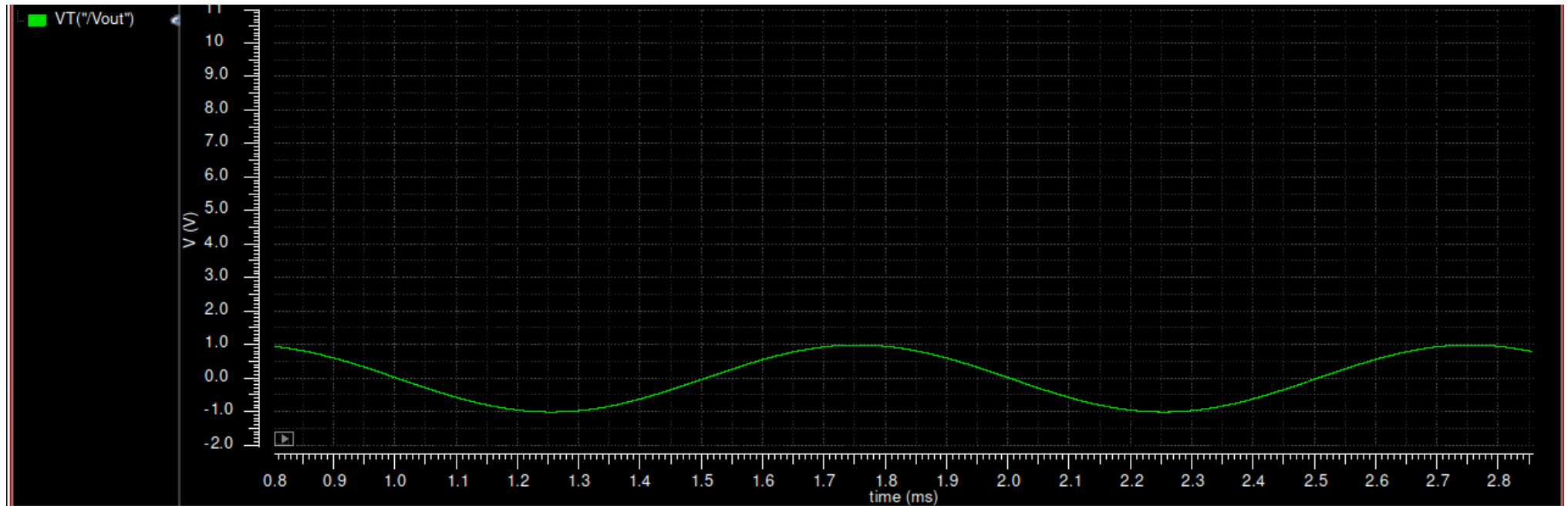
Without IC, loop will not start (DC operating point gets stuck)

Part 1: Schematic



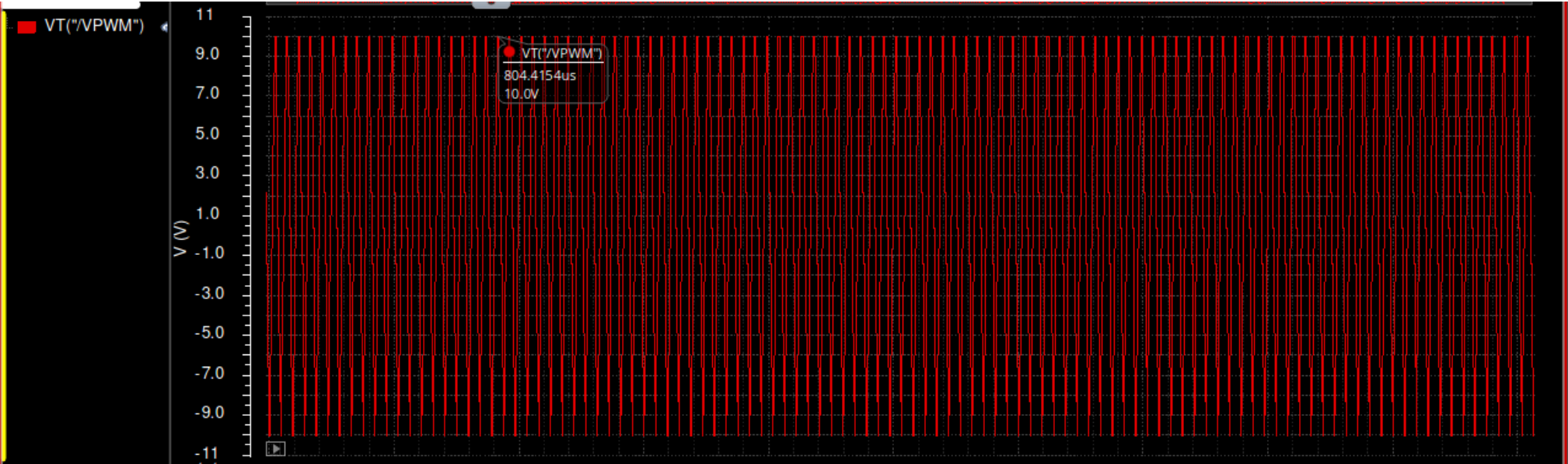
1st order feedback loop with limiting_diffamp, subtractor, hysteresis

Part 1: Vout Waveform (1kHz, 10uA)



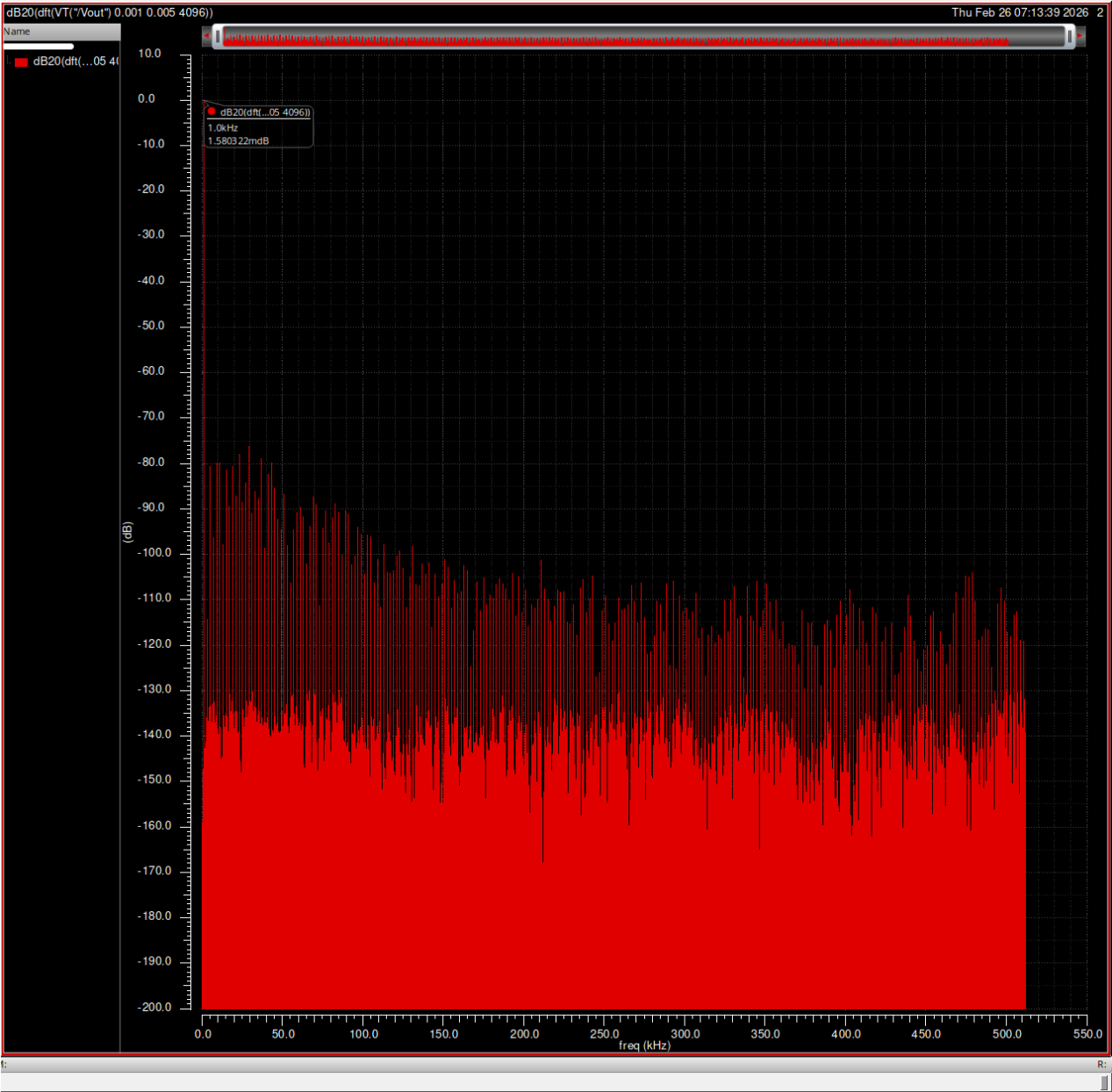
Clean 1V sine at output, transimpedance = $lin * R_{FB}$

Part 1: VPWM Zoomed



PWM switching at ~1MHz with duty cycle modulated by input signal

Part 1: Spectrum (1kHz, 10uA)



Dominant 1kHz peak at ~0 dB, PWM harmonics attenuated by LC filter

Part 1: Maximum Current Amplitude

Theoretical: $I_{IN,MAX} = V_{out,max} / R_{FB} = 10V / 100k = 100 \mu A$

Simulated: $I_{IN,MAX} = 74 \mu A$

Why lower than theoretical:

Integrator needs headroom for PWM triangle ripple on output (net4)

At 74 μA : $V_{out} = \pm 7.4V$, loop operates cleanly

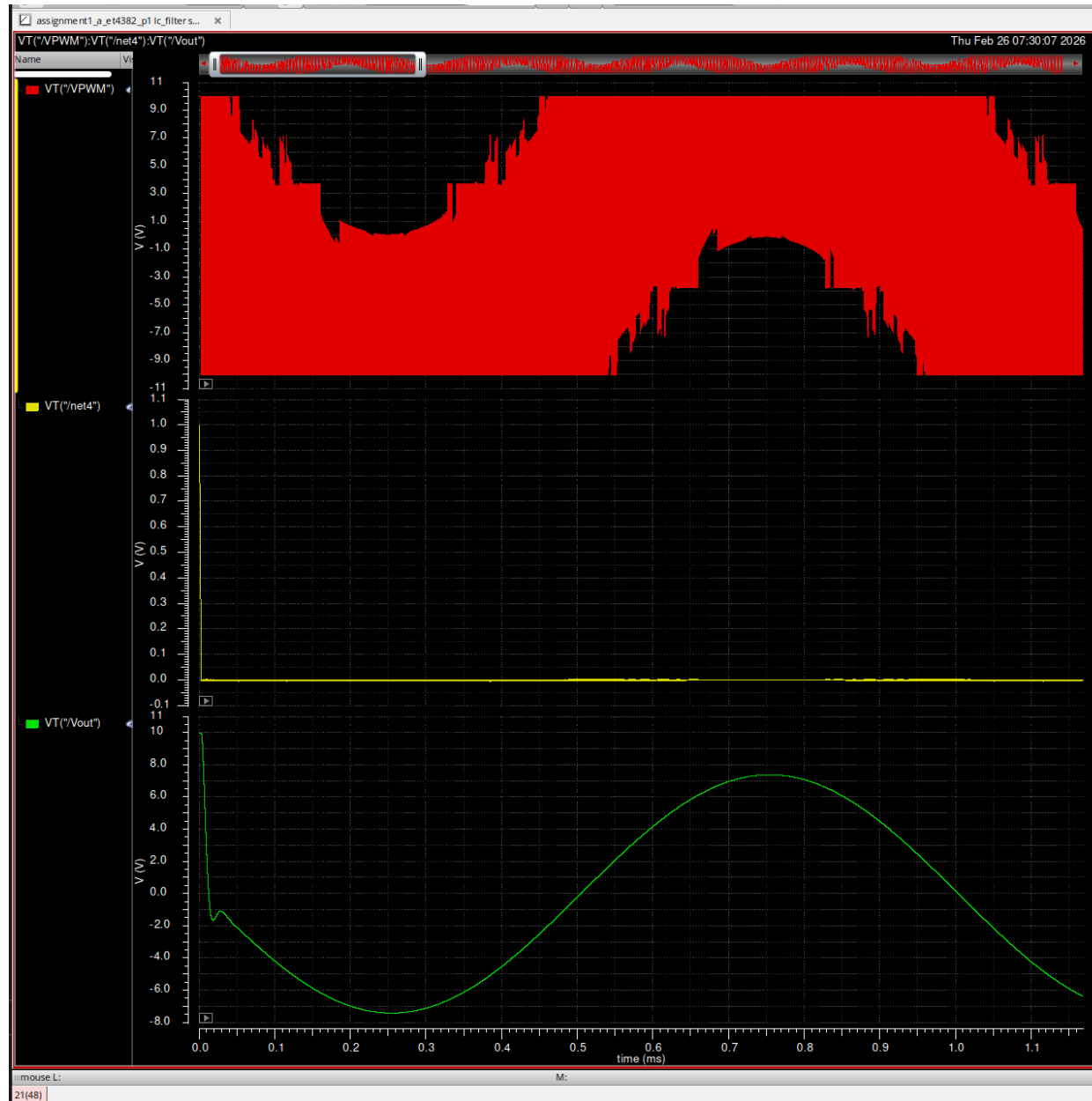
At 75 μA : integrator output clips at $\pm 2V$, VPWM loses switching, loop dies

Sweep results:

40 μA : $\pm 4V$ (clean) | 70 μA : $\pm 7V$ (clean)

74 μA : $\pm 7.4V$ (max) | 75 μA : saturated (loop dies)

Part 1: IIN,MAX = 74 uA



Maximum clean operation at 74 uA

Part 1: Clipping at 75 uA



Loop saturates: VPWM stuck, Vout drifts

Part 1: 5kHz Error Injection

Setup:

Verr (vsource, sine, 5kHz, 1V) placed between I12 sigout and VPWM node

Both LC filter and feedback path see the error

I13 amplitude = 0 (no input signal)

Result: 5kHz peak at -73.1 dB in output spectrum

Error suppression = 73 dB

Theoretical loop gain @ 5kHz:

$$KPWM = VPWM / (\pi * V_{clip}) = 10 / (\pi * 2) = 1.59$$

$$|Hint(5kHz)| = f_{UGB} / f = 6.37k / 5k = 1.27$$

$$|L(5kHz)| = 1.27 * 1.59 = 2.02 \text{ (~6 dB)}$$

$$\text{Predicted suppression: } 1 + |L| = 3 \rightarrow 9.5 \text{ dB}$$

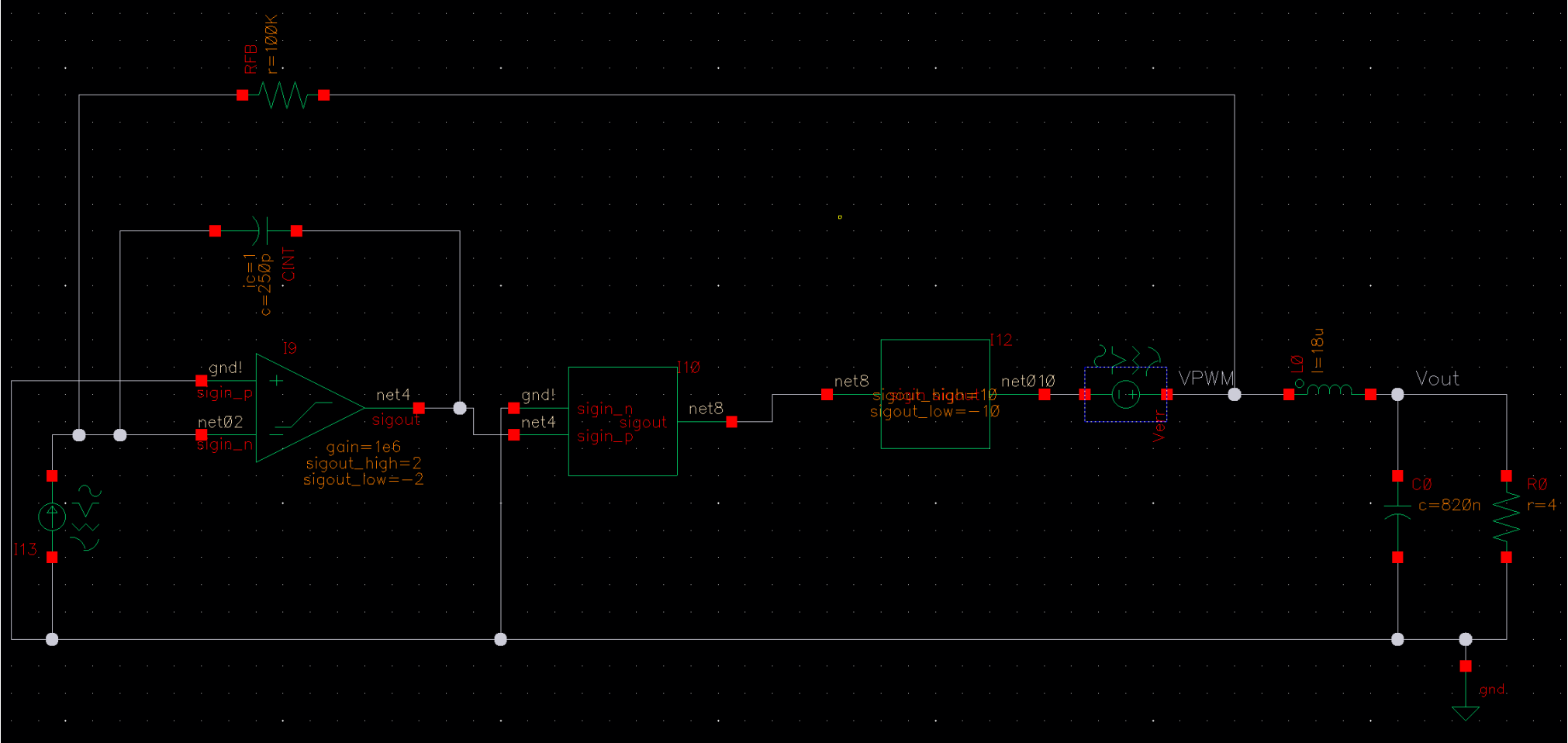
Measured 73 dB >> predicted 9.5 dB:

Self-oscillating loop has much higher effective KPWM

Triangle ripple is only ~0.2V pp (not +/-2V clipping limits)

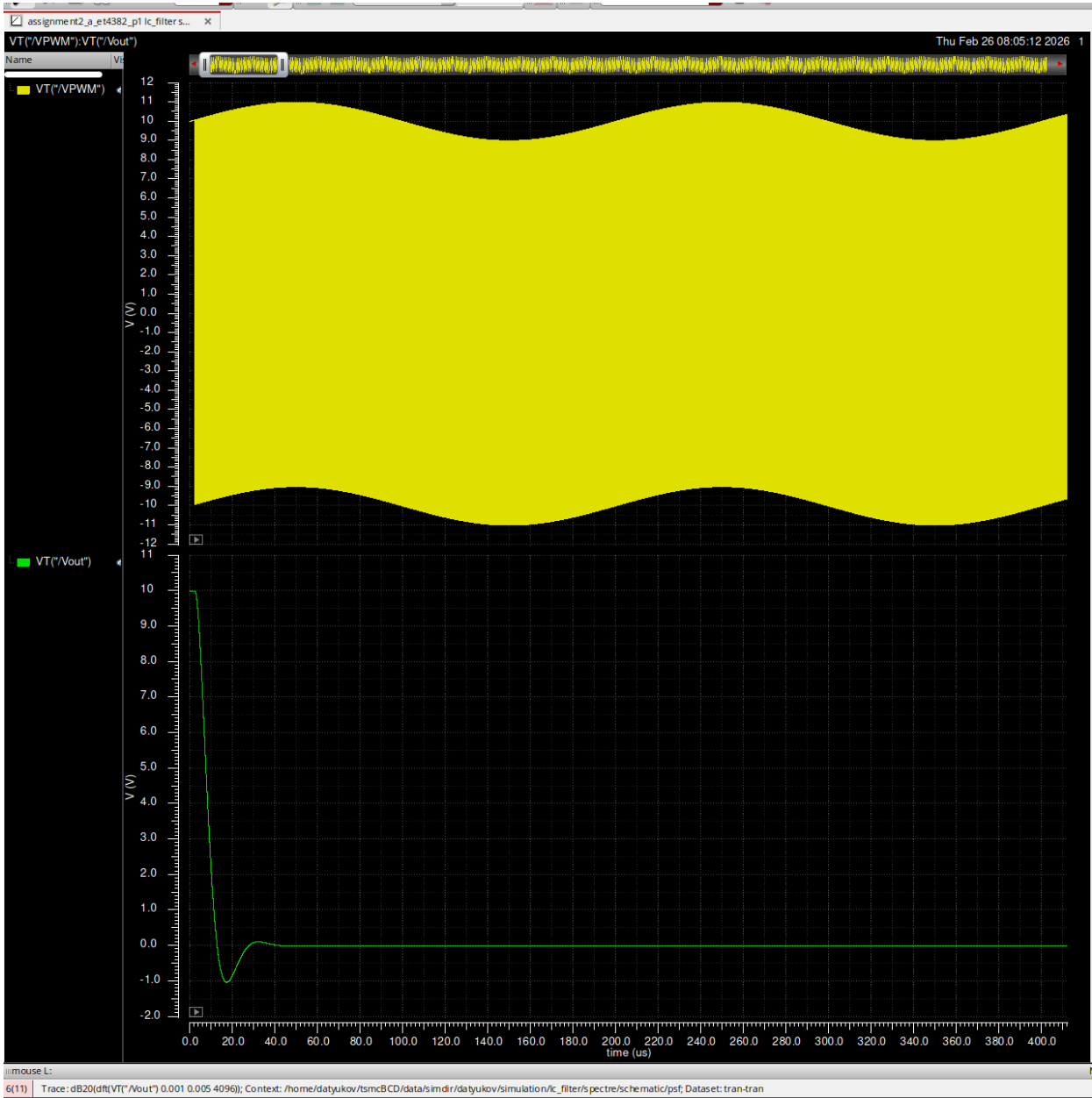
$$\text{Effective KPWM} = 10 / (\pi * 0.1) = 31.8$$

Part 1: Error Injection Schematic



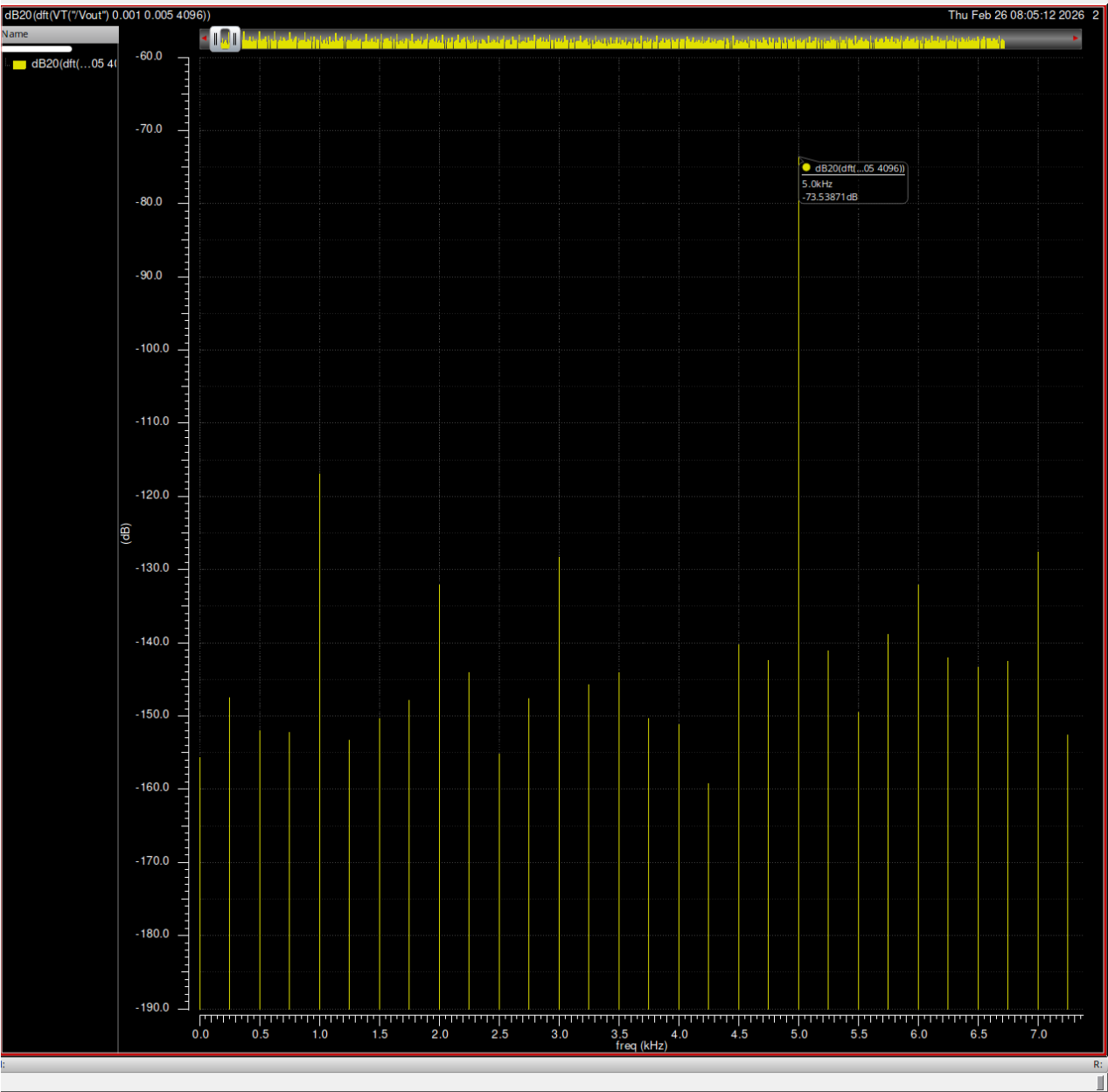
Verr placed between hysteresis output and VPWM node
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Part 1: Error Waveform



VPWM with 5kHz envelope, Vout settled near 0V
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Part 1: Error Spectrum



5kHz error suppressed to -73.1 dB at output

Part 2

2nd Order Feedback Loop

Part 2: Circuit Description

Added to 1st order loop:

2nd integrator I14 (limiting_diffamp, same settings as I9)

RFB2 = 100 kOhm from VPWM to I14 virtual ground (net03)

CINT2 = 250 pF (IC = 1V) on I14

RZERO = 500 kOhm in series with CINT on I9 (stabilizing zero)

10 pF cap on net03 to gnd (convergence fix)

Signal chain:

I9 sigout (net013) -> I10 subtractor sign_p

I14 sigout (net04) -> I10 subtractor sign_n (triangle reference)

Subtractor -> hysteresis -> VPWM

VPWM -> RFB -> I9, VPWM -> RFB2 -> I14, VPWM -> LC filter

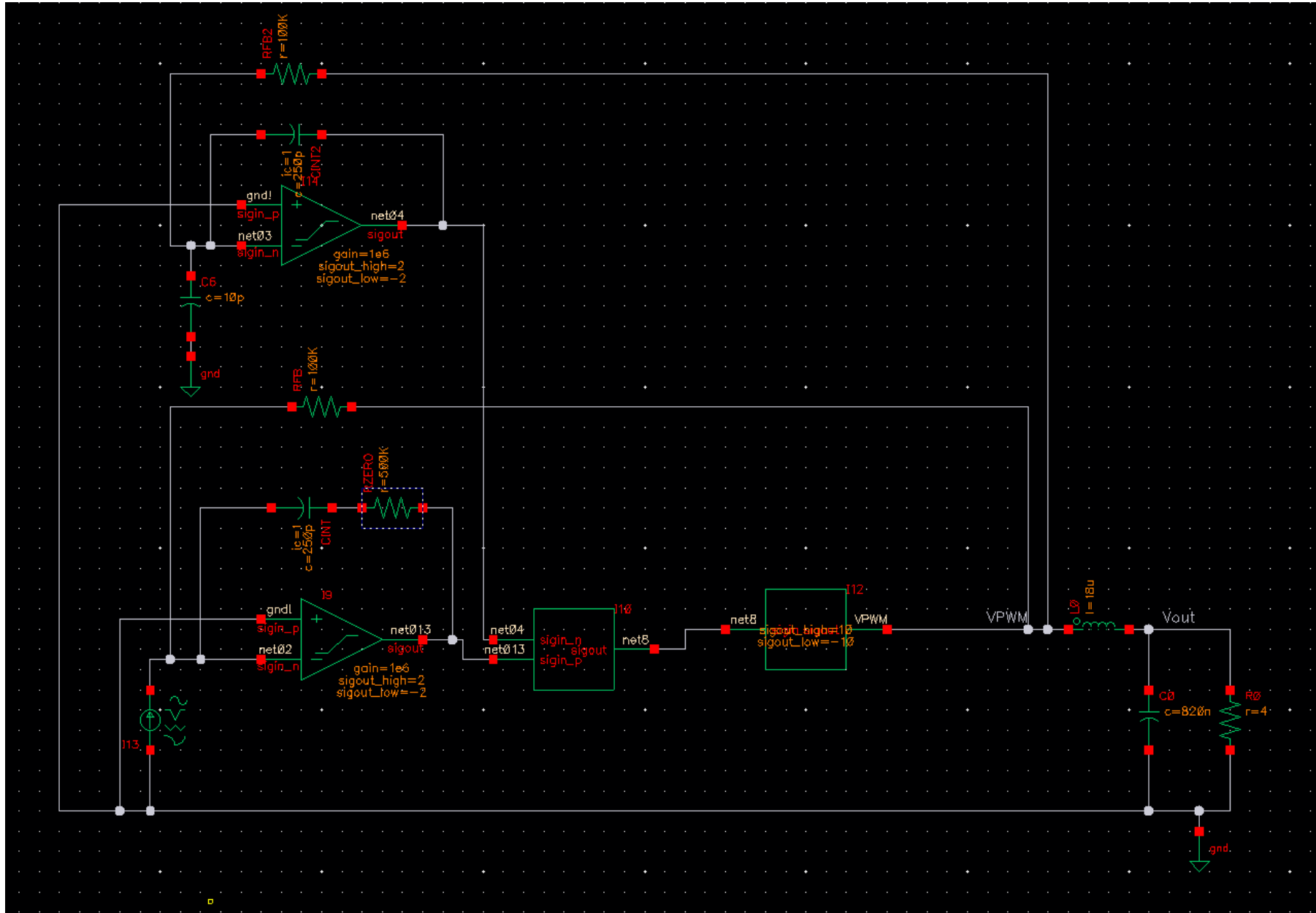
Zero frequency: $f_z = 1/(2\pi * 500k * 250p) = 1.27 \text{ kHz}$

Observation: loop self-oscillates at ~50 kHz (not 1 MHz)

Both integrators have identical time constants ($RFB * CINT = 25 \text{ us}$)

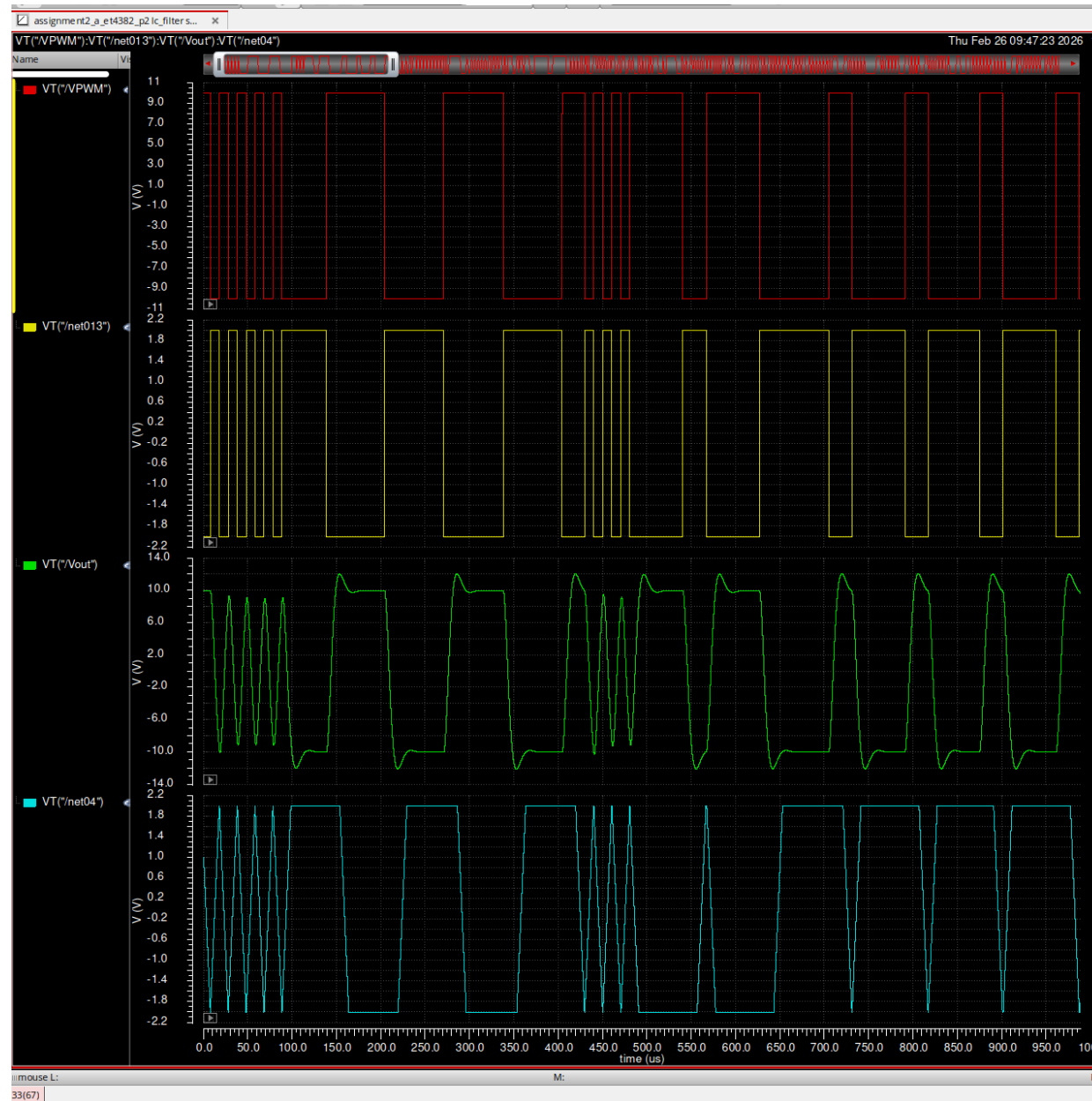
This causes the integrator poles to cancel in the loop transfer

Part 2: Schematic



2nd order feedback loop with two integrators and stabilizing zero

Part 2: Waveforms



VPWM, net013 (I9 out), Vout, net04 (I14 out / triangle reference)

Part 2: 5kHz Error Injection

Setup: same as Part 1 Step 4

Verr (5kHz, 1V) between I12 sigout and VPWM node

I13 amplitude = 0

Result: 5kHz peak at -6.4 dB in output spectrum

Error suppression = 6.4 dB

Much lower than Part 1's 73 dB because:

Identical integrator time constants ($C_{INT} = C_{INT2} = 250 \text{ pF}$)

cause the frequency-dependent terms to cancel

Loop acts as constant gain: $L = K_{PWM} * R_{ZERO}/R_{FB} = 1.59 * 5 = 8$

No integrating behavior -> minimal error suppression

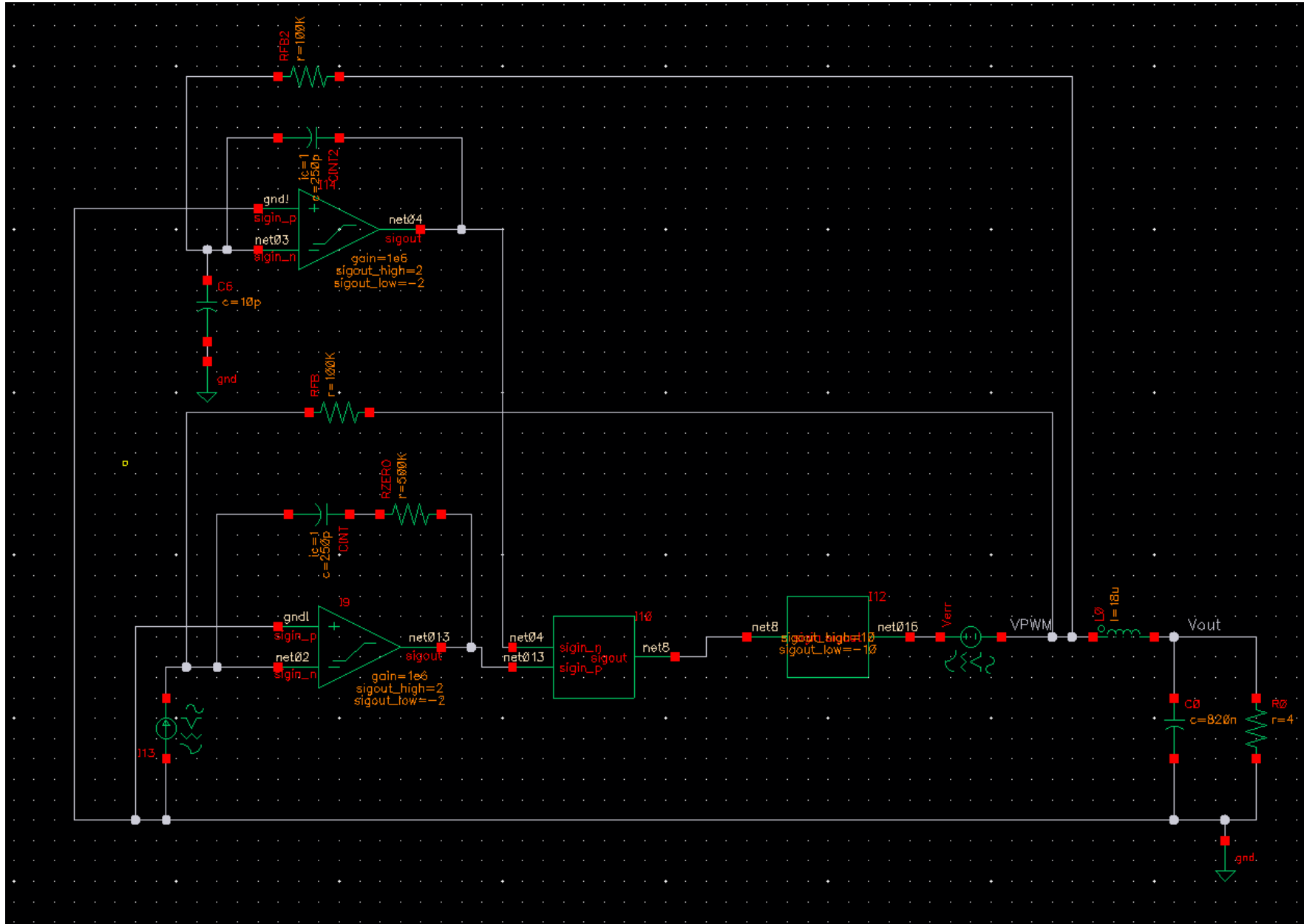
Theoretical suppression:

$|L| = K_{PWM} * R_{ZERO}/R_{FB} = 7.95$ (~18 dB)

Predicted: $20 * \log_{10}(1 + 7.95) = 19 \text{ dB}$

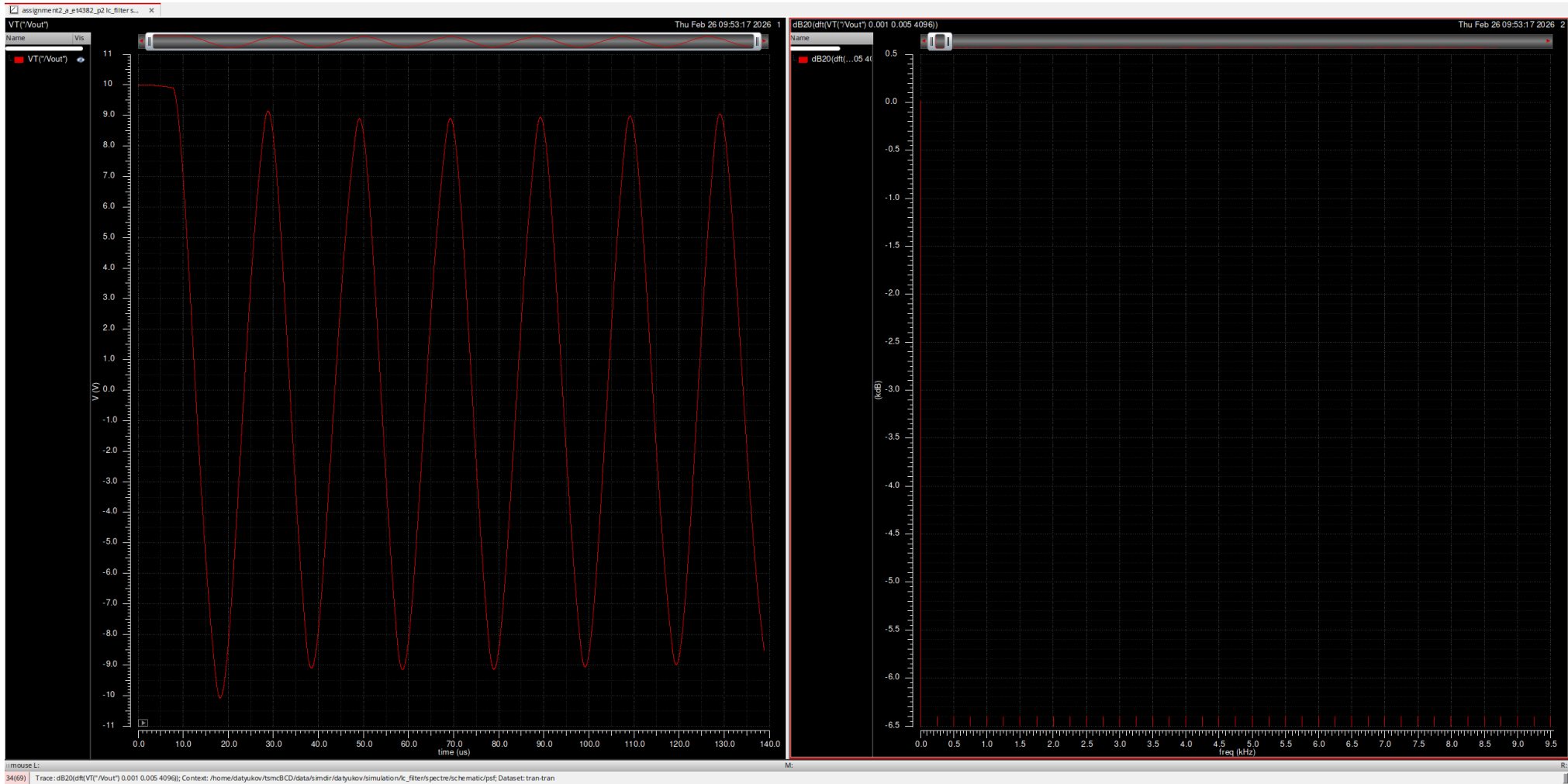
Measured: 6.4 dB (lower due to chaotic loop operation)

Part 2: Error Injection Schematic



Verr placed between hysteresis output and VPWM node

Part 2: Error Spectrum



5kHz error at -6.4 dB (only 6 dB suppression vs Part 1's 73 dB)
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Part 3

Clipping Comparison (FS + 10%)

Part 3: Setup & Results

Drive both loops at FS+10%:

$$I_{in} = 1.1 * I_{IN,MAX} = 1.1 * 74 \text{ uA} = 81 \text{ uA}$$

I13: sine, 1kHz, 81 uA

1st order loop result:

Vout saturates to -10V and stays there permanently

Integrator clips at -2V, VPWM locks to -10V, loop dies

Clean but fatal saturation - output locks to one rail

Once overdriven past $I_{IN,MAX}$, the loop cannot recover

2nd order loop result:

Vout oscillates chaotically between +/-12V with erratic swings

Both integrators clip and fight each other (integrator wind-up)

LC filter ringing causes overshoot beyond +/-10V (peaks to +/-14V)

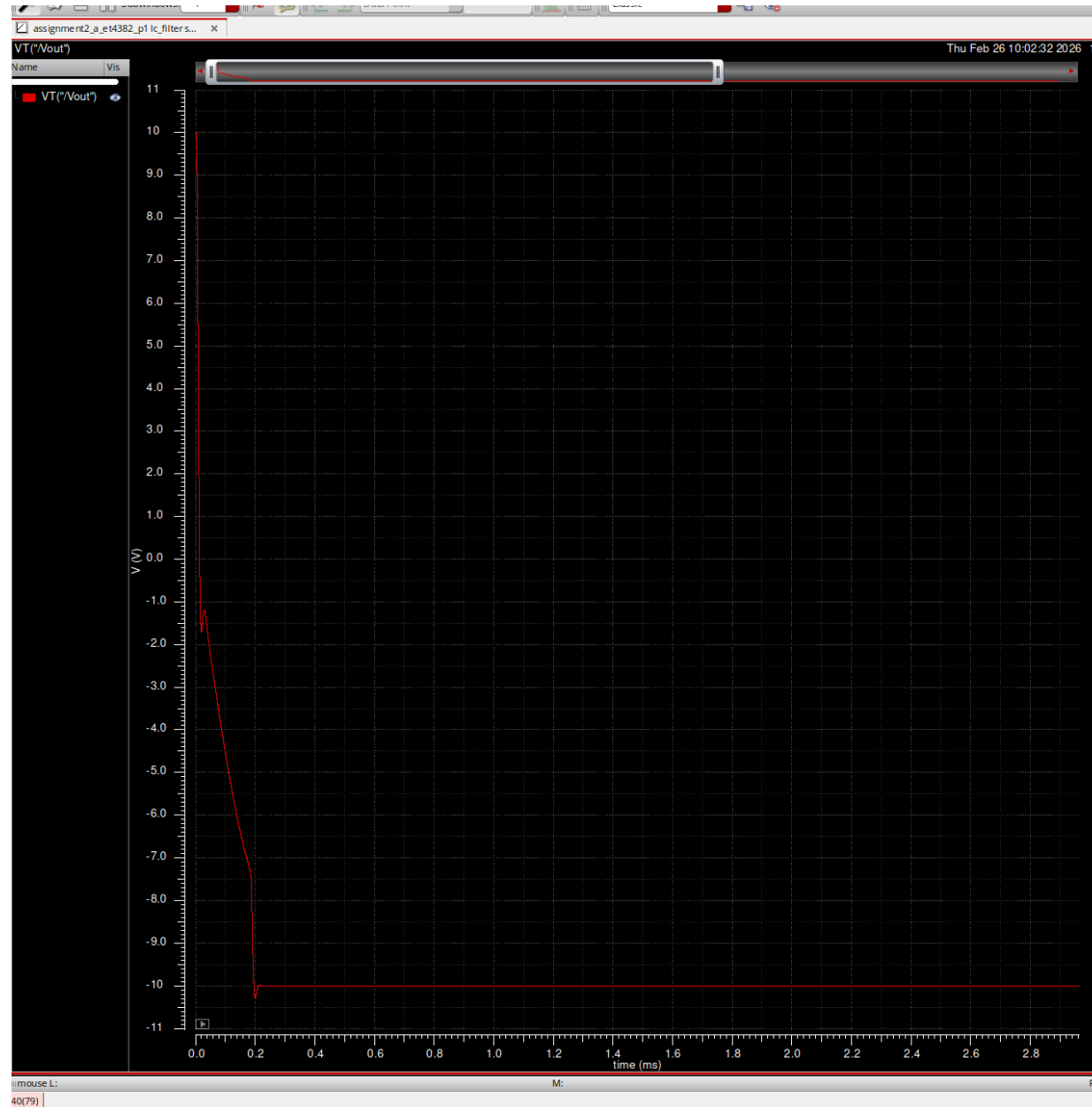
Loop never settles but also never fully locks up

Key difference:

1st order: fails gracefully (locks to rail, silent failure)

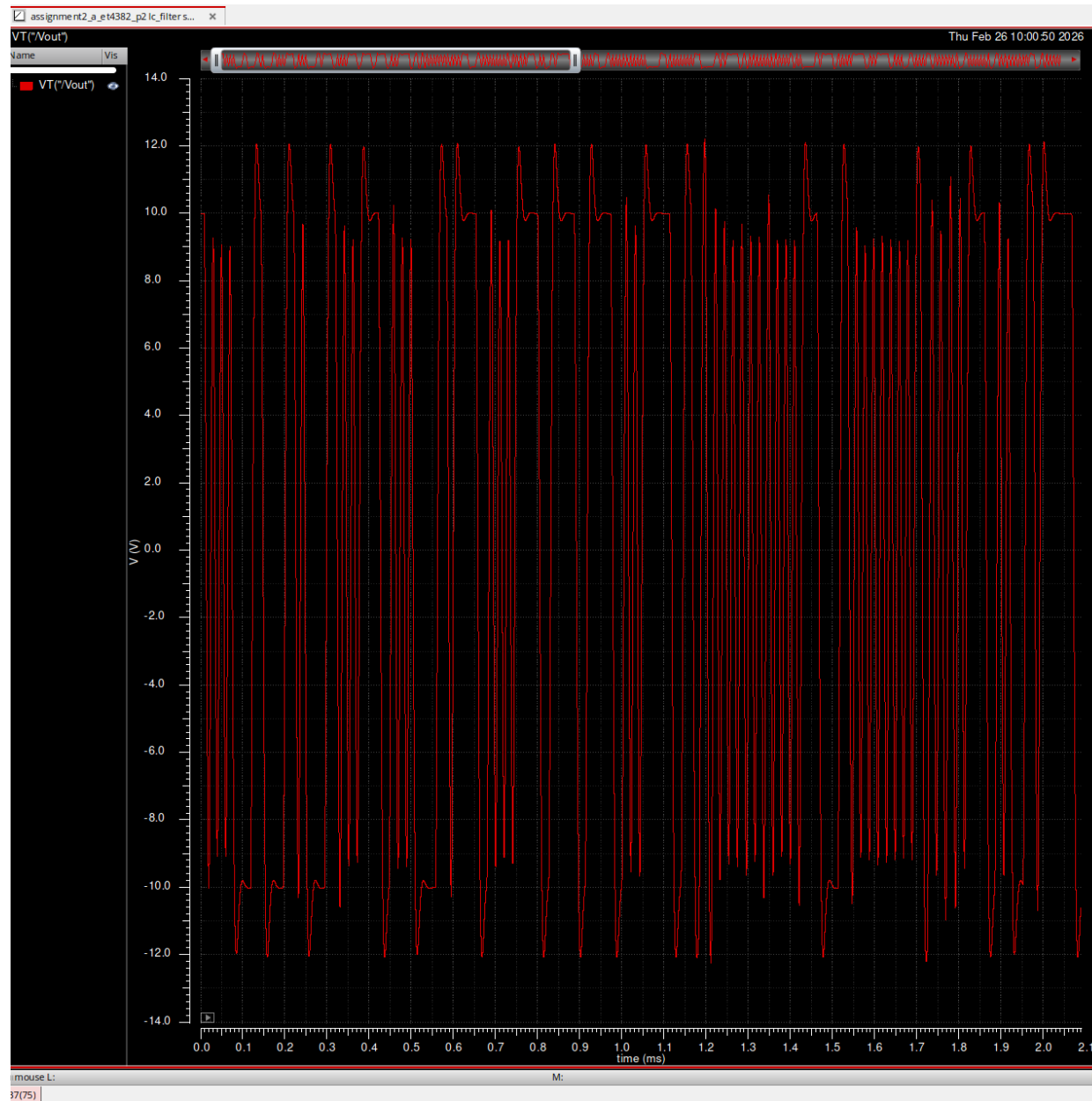
2nd order: fails chaotically (wind-up, unstable oscillation with overshoot)

Part 3: 1st Order Clipping (81 uA)



1st order loop: Vout saturates to -10V permanently

Part 3: 2nd Order Clipping (81 uA)



2nd order loop: chaotic oscillation +/-12V with integrator wind-up

Summary

Part 1: 1st Order Feedback Loop

Self-oscillating PWM at $\sim 1\text{MHz}$, $C_{INT} = 250\text{ pF}$, $R_{FB} = 100\text{ k}\Omega$

$I_{IN,MAX} = 74\text{ }\mu\text{A}$, transimpedance = $R_{FB} = 100\text{ k}\Omega$

Error suppression @ $5\text{kHz} = 73\text{ dB}$ (self-oscillating gain $\gg$ naturally-sampled)

Part 2: 2nd Order Feedback Loop

Added 2nd integrator (I_{14}) + $R_{ZERO} = 500\text{ k}\Omega$ stabilizing zero

Self-oscillation at $\sim 50\text{ kHz}$ (lower due to 2nd integrator dynamics)

Error suppression @ $5\text{kHz} = 6.4\text{ dB}$ (identical time constants limit performance)

Part 3: Clipping at FS+10% (81 μA)

1st order: V_{out} locks to -10V (clean, silent saturation)

2nd order: chaotic $\pm 12\text{V}$ oscillation (integrator wind-up)

2nd order harder to recover from clipping due to two integrators